

Program Overview

The program is a C program that takes in a file that has a list of commands. The program will handle these commands and print the output to the user's console.

Implementation

The program will define its own MaxHeap. The MaxHeap must have a list-based implementation that creates a priority queue. No pre-existing heap functions or modules can be used.

Email Definition and Priority

The MaxHeap should expect to take in an Email. An Email is an object with a category, subject, and date. The category field can only be "Boss", "Subordinate", "Peer", "ImportantPerson", and "OtherPerson". The MaxHeap should first try and sort the email by these categories with "Boss" being the most prioritized and "OtherPerson" being the least. In the event of a conflict, the date field should be used to place the email. The date field is defined in the format "MM-DD-YYYY" and is a string. The newest email will be prioritized over the older email in the event that they share the same category. The subject field is a string.

Commands

There are 4 commands that the program must handle.

EMAIL <sender_category>,<subject_line>,<date>	The email command will add an email to the MaxHeap adhering to the priority information in the previous section.
NEXT	Displays information for next email in MaxHeap in format: "Sender: Subject: Date:"
READ	Removes email from MaxHeap
COUNT	Displays count of unread emails

Input

The program assumes that the format of the test file adheres to the format declared above but does not assume that the commands are not nonsensical. There are a few rules that the program must follow:

- NEXT or READ are invalid if there are no emails in the MaxHeap
- NEXT is invalid without a READ command between
- READ is invalid without a NEXT command between

Example Run

- Sample Test File:

EMAIL Peer,Can you help me on this?,12-01-2024

EMAIL OtherPerson,Try our product,12-19-2024

EMAIL Boss,Important,12-20-2024

EMAIL Subordinate,How do I handle this?,12-25-2024

EMAIL ImportantPerson,Health Insurance Enrollment,12-31-2024

EMAIL Boss,Never Mind,01-03-2025

COUNT

NEXT

READ

NEXT

READ

COUNT

- Output of Sample Test File:

There are 6 emails to read.

Next email:

Sender: Boss

Subject: Never Mind

Date: 01-03-2025

Next email:

Sender: Boss

Subject: Important

Date: 12-20-2024

There are 4 emails to read.